

Purpose: To design an AI model that predicts whether a student will pass a test (Yes) or not pass a test (No).

Say: It classifies the input into two classes Yes and No. Also, calculate the accuracy of the classifier model and construct the confusion matrix for the model.

Activity Guidelines

- Let's assume you are testing your model on 1000 total test data.
- Out of which the actual values are 900 Yes and only 100 No (Unbalanced dataset).
- Let's assume that you have built a faulty model which, irrespective of any input, will give a prediction as Yes.
- Can you tell the classification accuracy of this model?

Step 1: Construct the Actual value vs Predicted value table

Actual value	Predicted Value

Consider 'Yes' as the positive class and 'No' as the negative class.

Step 2: Construct the confusion matrix.

Activity solution: Accuracy from Confusion matrix

So, the faulty model will predict all the 1000 input data as Yes.

Actual value	Predicted Value
Yes=900	Yes=1000
No=100	No=0

Consider 'Yes' as the positive class and 'No' as the negative class.

Construct the confusion matrix from the Actual vs Predicted table.

		Predicted Values	
		Yes	No
Actual Values	Yes	TP=	FN=
	No	FP=	TN=

Activity solution: Accuracy from Confusion matrix

		Predicted Values	
		Yes	No
Actual Values	Yes	TP=900	FN=0
	No	FP=100	TN=0

Step 3: Now calculate the accuracy from this matrix.

$$\begin{aligned}
 \text{Classification accuracy} &= \frac{\text{Correct predictions}}{\text{Total predictions}} \\
 &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}
 \end{aligned}$$

Step 4: Converting the accuracy to percentage: = %

$$\begin{aligned}
 \text{Classification accuracy} &= \frac{\text{Correct predictions}}{\text{Total predictions}} \\
 &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \\
 &= \frac{900}{900 + 0 + 100 + 0} = 0.9
 \end{aligned}$$

Converting the accuracy to percentage: $0.9 \times 100 \% = 90\%$

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So, the faulty model you made is showing an accuracy of 90%. Does this make sense?

So, *in cases of unbalanced data, we should use other metrics such as Precision, Recall or F1 score.*
Let's understand them one by one...

Precision from Confusion matrix

- **Precision** is the ratio of the total number of correctly classified positive examples and the total number of predicted positive examples.
- Precision = 0.843 means that when our model predicts a patient has heart disease, it is correct around 84% of the time.

		Predicted Values	
		1	0
Actual Values	1	TP=12	FN=06
	0	FP=04	TN=21

$$\text{Precision} = \frac{\text{Correct positive predictions}}{\text{Total positive predictions}}$$
$$= \frac{TP}{TP+FP}$$

Precision: where should we use it?

The metrics Precision is generally used for unbalanced datasets when dealing with the False Positives become important, and the model needs to reduce the FPs as much as possible.

Precision use case example

- For example, take the case of predicting a good day based on weather conditions to launch satellite.
- Let's assume a day with favorable weather condition is considered Positive class and a day with non-favorable weather condition is considered as Negative class.
- Missing out on predicting a good weather day is okay (low recall) but predicting the bad weather day (Negative class) as a good weather day (Positive class) to launch the satellite can be disastrous.
- So, in this case, the FPs need to be reduced as much as possible.





Recall from Confusion matrix

- The recall is the measure of our model correctly identifying True Positives
- Thus, for all the patients who actually have heart disease, recall tells us how many we correctly identified as having a heart disease. Recall = 0.86 tells us that out of the total patients who have heart disease 86% have been correctly identified.

$$\text{Recall} = \frac{\text{Correct positive predictions}}{\text{Total actual positive values}}$$

$$= \frac{TP}{TP+FN}$$

Do you know Recall is also called as Sensitivity or True Positive Rate?

Recall: Where we should we use it?

The metrics Recall is generally used for unbalanced dataset when dealing with the False Negatives become important and the model needs to reduce the FNs as much as possible.

Recall use case example

For example, for a covid-19 prediction classifier, let's consider detection of a covid-19 affected case as positive class and detection of covid-19 non-affected case as negative class.

- Imagine if a covid-19 affected person (Positive) is falsely predicted as non-affected of Covid-19 (Negative), the person if rely solely on the AI would not get any treatment and also may end up infecting many other persons.
- So, in this case, the FNs needs to be reduced as much as possible.
- Hence, Precision is a go-to metrics for this kind of use case.



F1 Score

- F1-Score provides a way to combine both precisions and recall into a single measure that captures both properties
- In those use cases, where the dataset is unbalanced, and we are unable to decide whether FP is more important or FN, we should use the F1 score as the suitable metric.

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Activity 4: Decide the appropriate metric to evaluate the AI model

Duration: 30 minutes

Purpose: To work with the given scenario and choose the most appropriate evaluation metric to evaluate their model.

Say: "Different evaluation metrics are used for evaluation in different scenarios and it is important that we realize how to choose the correct one."

Scenario: Flagging fraudulent transactions

- You have designed a model to detect any fraudulent transactions with credit card.
- You are testing your model with highly unbalanced dataset.
- What is the metric to be considered in this case?
- It is okay to classify a legit transaction as fraudulent — it can always be re-verified by passing through additional checks.
- But it is definitely not okay to classify a fraudulent transaction as legit (false negative).
- So here false negatives should be reduced as much as possible.
- Hence in this case, Recall is more important.
- For the given data, construct the confusion matrix.
- Calculate the recall from the confusion matrix.

Transaction ID	Actual value	Predicted Value
1	Non-Fraudulent	Non-Fraudulent
2	Non-Fraudulent	Fraudulent
3	Non-Fraudulent	Non-Fraudulent
4	Fraudulent	Non-Fraudulent
5	Fraudulent	Fraudulent
6	Non-Fraudulent	Non-Fraudulent
7	Fraudulent	Non-Fraudulent
8	Non-Fraudulent	Fraudulent
9	Non-Fraudulent	Non-Fraudulent
10	Non-Fraudulent	Non-Fraudulent

Fill the matrix based on the table given above.

		Predicted Values	
		Fraudulent	Non fraudulent
Actual Values	Fraudulent		
	Non-Fraudulent		

Activity solution: Decide the appropriate metric to evaluate the AI model

		Predicted Values	
		Fraudulent	Non fraudulent
Actual Values	Fraudulent	TP=01	FN=02
	Non-Fraudulent	FP=02	TN=05

Calculate the recall from the confusion matrix based on.

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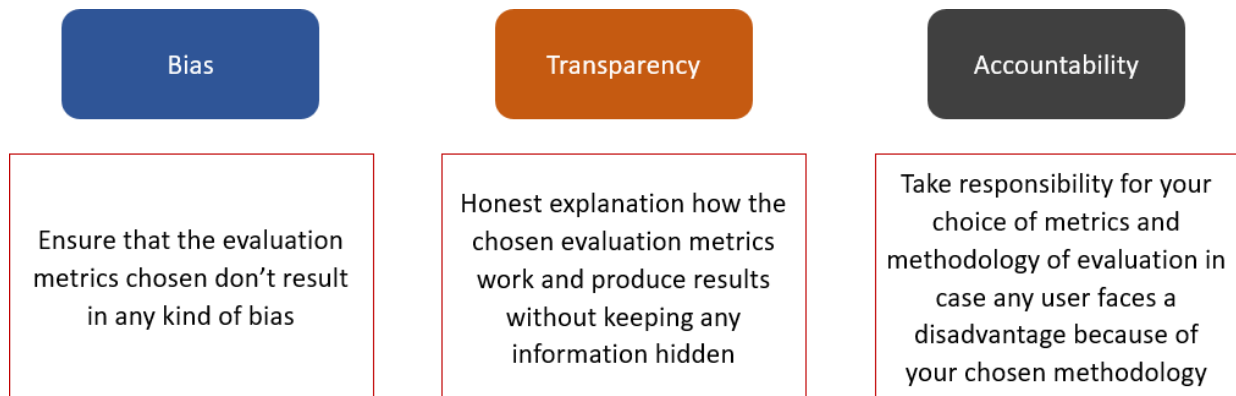
Write the formula for recall:

Calculate recall from the formula:

Are there any ethical concerns we need to keep in mind when performing model evaluation?

3.2 Ethical concerns around model evaluation

While evaluating an AI model, the following ethical concerns need to be kept in mind



Test Yourself

Choose the most appropriate answer for each question.

1. In a medical test for a rare disease, out of 1000 people tested, 50 actually have the disease while 950 do not. The test correctly identifies 40 out of the 50 people with the disease as positive, but it also wrongly identifies 30 of the healthy individuals as positive. What is the accuracy of the test?
A) 97%
B) 90%
C) 85%
D) 70%
2. A student solved 90 out of 100 questions correctly in a multiple-choice exam. What is the error rate of the student's answers?
A) 10%
B) 9%

D) 11%

3. In a spam email detection system, out of 1000 emails received, 300 are spam. The system correctly identifies 240 spam emails as spam, but it also marks 60 legitimate emails as spam. What is the precision of the system?
- A) 80%
- B) 70%
- C) 75%
- D) 90%
4. In a binary classification problem, a model predicts 70 instances as positive out of which 50 are actually positive. What is the recall of the model?
- A) 50%
- B) 70%
- C) 80%
- D) 100%
5. In a sentiment analysis task, a model correctly predicts 120 positive sentiments out of 200 positive instances. However, it also incorrectly predicts 40 negative sentiments as positive. What is the F1 score of the model?
- A) 0.8
- B) 0.75
- C) 0.72
- D) 0.82
6. A medical diagnostic test is designed to detect a certain disease. Out of 1000 people tested, 100 have the disease, and the test identifies 90 of them correctly. However, it also wrongly identifies 50 healthy people as having the disease. What is the precision of the test?
- A) 90%
- B) 80%
- C) 70%
- D) 60%
7. A teacher's marks prediction system predicts the marks of a student as 75, but the actual marks obtained by the student are 80. What is the absolute error in the prediction?
- A) 5
- B) 10
- C) 15
- D) 20

8. The goal when evaluating an AI model is to:
- A) Maximize error and minimize accuracy
 - B) Minimize error and maximize accuracy
 - C) Focus solely on the number of data points used
 - D) Prioritize the complexity of the model
9. A high F1 score generally suggests:
- A) A significant imbalance between precision and recall
 - B) A good balance between precision and recall
 - C) A model that only performs well on specific data points
 - D) The need for more training data
10. How is the relationship between model performance and accuracy described?
- A) Inversely proportional
 - B) Not related
 - C) Directly proportional
 - D) Randomly fluctuating

Reflection Time:

Q1. What will happen if you deploy an AI model without evaluating it with known test set data? Q2.

Do you think evaluating an AI model is that essential in an AI project cycle?

Q3. Explain train-test split with an example.

Q4. "Understanding both error and accuracy is crucial for effectively evaluating and improving AI models." Justify this statement.

Q5. What is classification accuracy? Can it be used all times for evaluating AI models?

Assertion and reasoning-based questions:

Q1. Assertion: Accuracy is an evaluation metric that allows you to measure the total number of predictions a model gets right.

Reasoning: The accuracy of the model and performance of the model is directly proportional, and hence better the performance of the model, the more accurate are the predictions.

Choose the correct option:

- (a) Both A and R are true and R is the correct explanation for A
- (b) Both A and R are true and R is not the correct explanation for A
- (c) A is True but R is False
- (d) A is false but R is True

Q2. Assertion: The sum of the values in a confusion matrix's row represents the total number of instances for a given actual class.

Reasoning: This enables the calculation of class-specific metrics such as precision and recall, which are essential for evaluating a model's performance across different classes.

Choose the correct option:

- (a) Both A and R are true and R is the correct explanation for A
- (b) Both A and R are true and R is not the correct explanation for A
- (c) A is True but R is False
- (d) A is false but R is True

Case study-based questions:

Q1. Identify which metric (Precision or Recall) is to be used in the following cases and why?

- a) Email Spam Detection
- b) Cancer Diagnosis
- c) Legal Cases (Innocent until proven guilty)
- d) Fraud Detection
- e) Safe Content Filtering (like Kids YouTube)

Q2. Examine the following case studies. Draw the confusion matrix and calculate metrics such as accuracy, precision, recall, and F1-score for each one of them.

a. Case Study 1:

A spam email detection system is used to classify emails as either spam (1) or not spam (0). Out of 1000 emails:

- True Positives (TP): 150 emails were correctly classified as spam.
- False Positives (FP): 50 emails were incorrectly classified as spam.
- True Negatives (TN): 750 emails were correctly classified as not spam.
- False Negatives (FN): 50 emails were incorrectly classified as not spam.
-

b. Case Study 2:

A credit scoring model is used to predict whether an applicant is likely to default on a loan (1) or not (0). Out of 1000 loan applicants:

- True Positives (TP): 90 applicants were correctly predicted to default on the loan.
- False Positives (FP): 40 applicants were incorrectly predicted to default on the loan.
- True Negatives (TN): 820 applicants were correctly predicted not to default on the loan.

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- False Negatives (FN): 50 applicants were incorrectly predicted not to default on the loan.

Calculate metrics such as accuracy, precision, recall, and F1-score.

c. Case Study 3:

A fraud detection system is used to identify fraudulent transactions (1) from legitimate ones (0). Out of 1000 transactions:

- True Positives (TP): 80 transactions were correctly identified as fraudulent.
- False Positives (FP): 30 transactions were incorrectly identified as fraudulent.
- True Negatives (TN): 850 transactions were correctly identified as legitimate.
- False Negatives (FN): 40 transactions were incorrectly identified as legitimate.

d. Case Study 4:

A medical diagnosis system is used to classify patients as having a certain disease (1) or not having it (0). Out of 1000 patients:

- True Positives (TP): 120 patients were correctly diagnosed with the disease.
- False Positives (FP): 20 patients were incorrectly diagnosed with the disease.
- True Negatives (TN): 800 patients were correctly diagnosed as not having the disease.
- False Negatives (FN): 60 patients were incorrectly diagnosed as not having the disease.

e. Case Study 5:

An inventory management system is used to predict whether a product will be out of stock (1) or not (0) in the next month. Out of 1000 products:

- True Positives (TP): 100 products were correctly predicted to be out of stock.
- False Positives (FP): 50 products were incorrectly predicted to be out of stock.
- True Negatives (TN): 800 products were correctly predicted not to be out of stock.
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- False Negatives (FN): 50 products were incorrectly predicted not to be out of stock.